

Article

Improved Genetic Algorithm-Based Path Planning for Multi-Vehicle Pickup in Smart Transportation

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Highlights

What are the main findings?

- This study proposes an Improved Genetic Algorithm (IGA) specifically designed for multi-vehicle path planning in large-scale and dynamic ride-hailing applications, which includes vehicle capacity constraints, incomplete station connectivity, and edge-based passenger distribution.
- Comparative experiments on four benchmark simulation environments demonstrate that the proposed IGA significantly reduces both maximum and average sub-path lengths, while improving computational efficiency relative to the classical MSTC* approach.

What is the implication of the main finding?

- The proposed IGA framework effectively addresses the operational challenges of multi-vehicle coordination in complex and intelligent transportation systems, offering a scalable and efficient solution for real-time vehicle path planning with shared-ride constraints.
- These findings provide methodological support for the development of intelligent dispatching strategies in intelligent transportation systems, with potential application in dynamic and resource-constrained urban mobility scenarios.



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Abstract

With the rapid development of intelligent transportation systems and online ride-hailing platforms, the demand for promptly responding to passenger requests while minimizing vehicle idling and travel costs has grown substantially. This paper addresses the challenges of suboptimal vehicle path planning and partially connected pickup stations by formulating the task as a Capacitated Vehicle Routing Problem (CVRP). We propose an Improved Genetic Algorithm (IGA)-based path planning model designed to minimize total travel distance while respecting vehicle capacity constraints. To handle scenarios where certain pickup points are not directly connected, we integrate graph-theoretic techniques to ensure route continuity. The proposed model incorporates a multi-objective fitness function, a rank-based selection strategy with adjusted weights, and Dijkstra-based path estimation to enhance convergence speed and global optimization performance. Experimental evaluations on four benchmark maps from the Carla simulation platform demonstrate that the proposed approach can rapidly generate optimized multi-vehicle path planning solutions and effectively coordinate pickup tasks, achieving significant improvements in both route quality and computational efficiency compared to traditional methods.

Keywords: improved genetic algorithm; intelligent transportation systems; multi-vehicle coordination; path planning; task allocation; capacitated vehicle routing problem

1. Introduction

The rapid advancement of intelligent transportation systems and the widespread adoption of online ride-hailing platforms have significantly reshaped urban mobility patterns [1]. As cities grow more densely populated, the demand for efficient, flexible, and high-quality transportation services has surged. Ride-hailing platforms have emerged as a prominent solution, offering advantages in convenience, comfort, and personalized service. However, this evolution also brings forth new challenges for intelligent transportation systems, particularly in balancing operational efficiency with user satisfaction [2].

A critical challenge in large-scale ride-hailing operations is the optimization of vehicle task allocation and path planning under realistic urban constraints [3]. Traditional path planning models often result in high idle mileage, inefficient resource allocation, and limited adaptability when responding to dynamic passenger demand. Many of these approaches are built on simplified assumptions—such as complete connectivity and uniform travel distances—that fail to capture the complexity of actual urban environments, where network disruptions, uneven accessibility, and incomplete station connectivity are common. Recent studies have underscored the importance of improving system responsiveness and overall transportation efficiency by accounting for these real-world operational factors [4,5]. These insights align closely with the objective of this work, which seeks to develop a more realistic and adaptive path planning framework for multi-vehicle coordination in urban ride-hailing scenarios.

To address these limitations, this study focuses on the multi-vehicle pickup path planning problem in the context of intelligent urban transportation. The problem is modeled as a realistic variant of the CVRP [6], extended to incorporate critical urban constraints including vehicle capacity, road network topology, and non-fully connected pickup points. This formulation enables more accurate modeling of operational complexity in dense urban ride-hailing systems.

To solve the proposed problem effectively, we introduce the IGA tailored for real-time, large-scale deployment. Enhancements to the traditional genetic algorithm include a customized multi-objective fitness function, an adaptive weight selection strategy, and a Dijkstra-based path estimation mechanism. These modifications improve convergence speed and solution quality while ensuring compatibility with urban graph structures and operational constraints. The specific workflow is illustrated in Figure 1.

The primary contributions of this work are as follows:

- We propose a novel multi-vehicle path planning model tailored for multi-passenger pickup scenarios in urban intelligent transportation systems. By extending the conventional CVRP to incorporate realistic constraints such as urban road topology and partially connected pickup stations, the proposed model significantly improves practical applicability.
- We design the IGA with a focus on accelerating convergence and enhancing global optimization capability, making it more suitable for large-scale and dynamic ride-hailing applications.
- We validate our approach using extensive simulations on four diverse maps in the Carla simulation platform. Results demonstrate significant improvements in routing quality and computational performance compared to traditional methods.

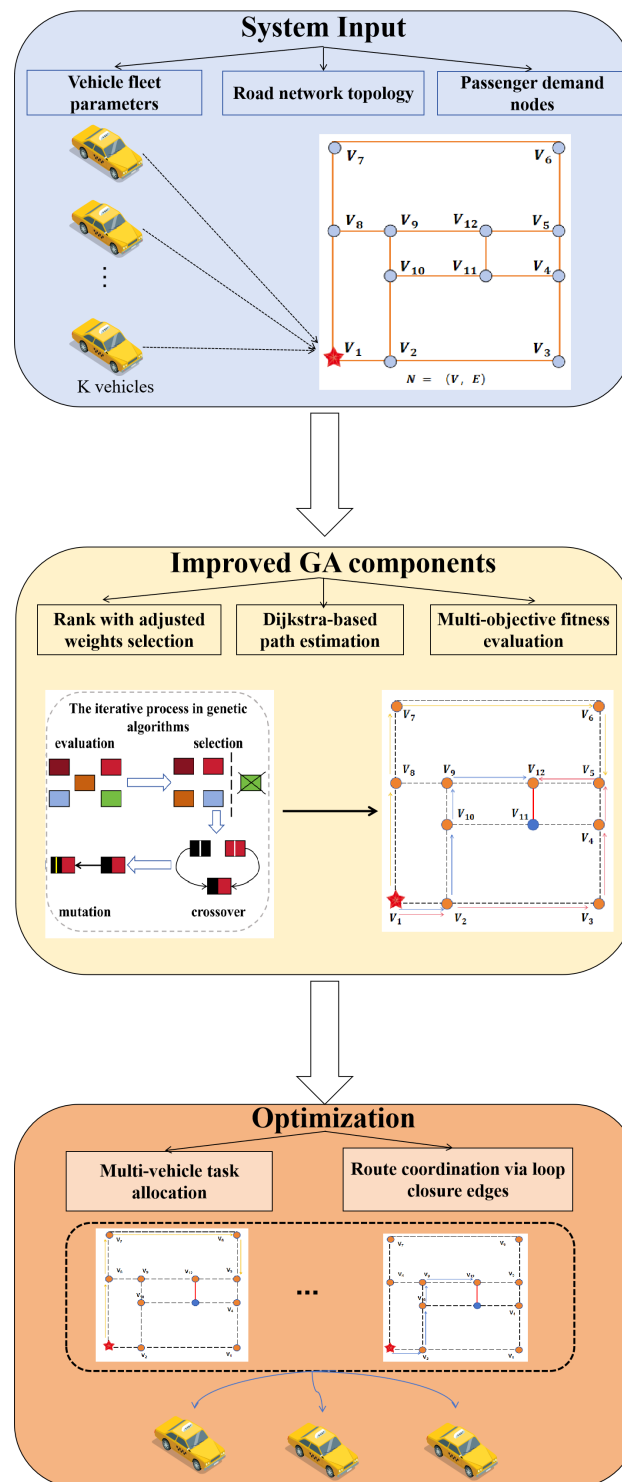


Figure 1. Overall framework of the IGA for multi-vehicle pickup planning. The system (1) constructs the road network graph from geospatial data, (2) initializes vehicle routes with capacity constraints, (3) optimizes paths through selection, crossover, and mutation operations with Dijkstra-based distance estimation.

The remainder of this paper is organized as follows. Section 2 reviews the related research results, Section 3 introduces the mathematical model of the vehicle pickup path planning problem, Section 4 elaborates the process of IGA, Section 5 conducts the experimental design and result analysis, Section 6 summarizes the contribution of this paper and looks forward to the future research direction.

2. Related Research

2.1. Traditional Path Planning Methods

Traditional path planning methods have long been the foundation of both academic research and practical applications in robotics and transportation. These approaches are typically divided into five main categories: graph-based, sampling-based, gradient-based, optimization-based, and interpolated curve methods. Among them, graph-based methods such as Dijkstra's and A* algorithms [7–9] are the most well-known and widely used. While these methods are effective for finding feasible paths, they often yield suboptimal or discontinuous trajectories, making them less suitable for complex, dynamic, or multi-vehicle scenarios that demand smooth and coordinated path planning.

Recent work in sampling-based methods, such as Rapidly-exploring Random Trees (RRT) [10] and Probabilistic Roadmaps (PRM) [11], have aimed to improve path continuity by probabilistically exploring the space. However, these methods can be computationally expensive and are less efficient in high-dimensional, time-sensitive applications. In contrast, gradient-based methods leverage optimization techniques to iteratively improve the path [12]. These methods are often used in continuous environments, but they struggle with large-scale multi-vehicle coordination problems.

2.2. Ride-Sharing and Multi-Vehicle Task Allocation

In recent years, the increasing complexity of urban transportation networks and the rapid growth of ride-sharing services have stimulated significant research interest in coordinated multi-vehicle task allocation. Early contributions in this area primarily focused on user behavior modeling and system-level optimization. For example, Mitropoulos et al. [13] proposed a tri-objective optimization model that simultaneously addresses user satisfaction, travel time, and operational costs in ride-sharing systems.

More recent studies have shifted toward algorithmic approaches designed for dynamic and large-scale environments. Bongiovanni et al. [14] introduced a binary multi-objective ensemble method to mitigate passenger flow distortion in carpooling platforms, thereby improving rider–driver matching efficiency. Chen et al. [15] investigated multi-agent reinforcement learning techniques for adaptive ride-matching, enabling more responsive and flexible dispatching strategies.

Despite these advancements, many approaches still face challenges in handling fine-grained spatial constraints, such as disconnected road edges, and in modeling physical feasibility under vehicle capacity restrictions. Furthermore, dynamic methods frequently require substantial online data and extended training periods, which limit their robustness for time-critical or resource-constrained deployments. Existing formulations also tend to emphasize either operational efficiency or equity but rarely achieve both within a unified framework.

2.3. Genetic Algorithms for Path Planning

Genetic Algorithms (GA) have proven effective for solving various forms of vehicle path planning problems due to their global search capabilities and flexible encoding strategies [16,17]. Foundational contributions by John Holland [18] and David E. Goldberg [19] established the theoretical underpinnings of genetic algorithms, laying the groundwork for their applications in optimization and machine learning, including path planning. El Moudni and Alimi [20] proposed a dynamic GA-based approach, demonstrating its effectiveness in real-time obstacle-avoidance scenarios.

Recent studies have explored genetic algorithms (GA) in various transportation optimization contexts. For instance, Taghavi et al. [21] proposed a multi-objective GA framework for the bus terminal location problem, which integrates data envelopment analysis

(DEA) to simultaneously optimize terminal placement efficiency and operational performance. Their approach demonstrates how GAs can effectively handle multi-criteria decision-making in large-scale transportation systems. In the context of capacitated vehicle routing, Wang and Lu [22] developed a hybrid genetic algorithm combining global search capabilities of GA with local search heuristics to improve both solution quality and computational efficiency. Their method proved especially effective in reducing the total distance traveled while maintaining strict adherence to vehicle capacity constraints. These contributions underscore the versatility of GAs in addressing both spatial facility planning and complex routing problems under real-world constraints.

Nevertheless, existing GA-based approaches commonly exhibit three key limitations: (1) limited scalability when extended to high-density multi-vehicle contexts with overlapping routes, (2) insufficient treatment of real-world constraints such as disconnected pickup points or loop closure requirements, and (3) a lack of coordination mechanisms across vehicle tasks, often resulting in unbalanced workload distribution. Furthermore, many implementations rely on Euclidean or heuristic distance metrics rather than realistic road-network-based cost estimation, limiting their applicability in urban environments.

3. Modeling the Multi-Vehicle Collaborative Path Planning Problem

To formalize the multi-vehicle collaborative passenger pickup task, we construct a mathematical model based on the CVRP, incorporating realistic road network topology, vehicle constraints, and spatial cost estimation derived from geospatial data.

3.1. Scenario Assumptions for Pickup Path Planning

We model the multi-vehicle path planning problem as a Capacitated Vehicle Routing Problem (CVRP), which effectively handles vehicle capacity limitations while enabling coordinated path planning. As shown in Figure 2, we utilize a simulated road map as prior knowledge to emulate a realistic transportation network, where K vehicles are dispatched to pick up passengers distributed throughout the city. Unlike classical formulations where pickup points are located at network nodes, our model assumes that passengers are positioned along road edges, reflecting practical scenarios in which passengers wait by the roadside rather than at intersections.

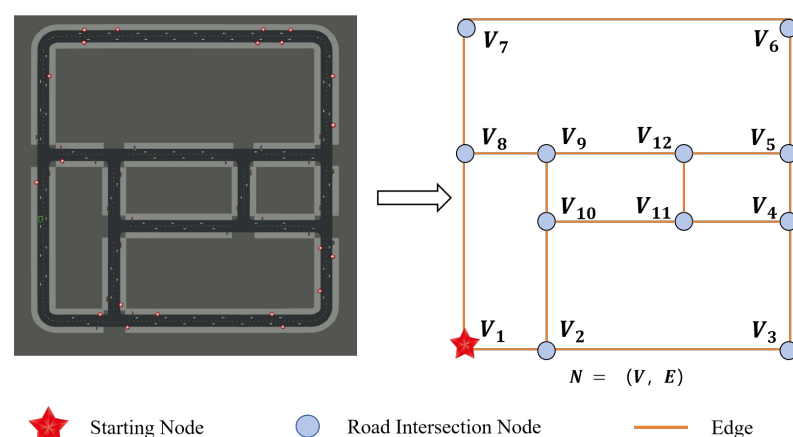


Figure 2. Construction of the road network topology graph $N = (V, E)$. Vertices $V = \{v_1, \dots, v_n\}$ represent intersections (e.g., v_7 – v_{12} shown), while edges E denote traversable roads. Passengers are distributed along edges rather than nodes. Edge weights are computed using the Haversine formula from (lat, lon) data, forming the basis for CVRP optimization with K capacity-constrained vehicles.

Specifically, due to the fact that passenger pickup locations are modeled along road segments rather than discrete nodes, the system is less sensitive to small errors or noise in GPS data. This edge-based approach reduces the impact of inaccuracies, such as minor

localization errors or signal noise, ensuring more stable performance even when the GPS data is imperfect. This characteristic enhances the robustness of the system, making it more reliable in real-world environments where noise and inaccuracies are common.

While this study assumes that all vehicles start from a common depot and that passengers agree to shared rides, the proposed edge-based CVRP formulation can be extended to more heterogeneous and dynamic pickup conditions. For example, time-window constraints can be incorporated into the fitness function to handle scheduled pickups. Dynamic vehicle entry or exit, as well as delayed or newly arriving pickup requests, can be addressed by adapting the chromosome encoding and evolutionary operators to allow variable fleet sizes and online task insertion. These extensions would enable real-time re-planning and improve the applicability of the method to real-world, on-demand ride-hailing and dispatching scenarios with continuously changing task information.

3.2. Mathematical Model

The multi-vehicle passenger pickup path planning problem is modeled as follows:

- All vehicles start from the same initial location.
- The number of vehicles is denoted by K .
- Passenger pickup requests are located along edges in the road network.
- Vehicles are initially unloaded.
- All passengers agree to shared rides.
- Vehicles are represented using a simplified kinematic model that operates along the road network graph. Dynamic aspects such as acceleration, braking forces, and tire dynamics are not explicitly modeled, as the focus of this work is on high-level path planning and task allocation.

We denote the set of nodes as $V = \{v_1, v_2, \dots, v_n\}$, where v_1 represents the depot. The edge set is $E = \{e_1, e_2, \dots, e_m\}$, and passenger requests are located along these edges. Let $R = \{r_1, r_2, \dots, r_q\}$ be the set of pickup requests, each associated with a position on some edge $e_{ij} \in E$ and a passenger count m_r .

Constraint 1: Vehicle Capacity Constraint

The total number of passengers picked up across all requests must not exceed the combined capacity of the vehicle fleet:

$$\sum_{r \in R} m_r \leq M \cdot K \quad (1)$$

In addition, each vehicle must not exceed its individual capacity:

$$\sum_{r \in R_k} m_r \leq M, \quad \forall k = 1, \dots, K \quad (2)$$

where $R_k \subseteq R$ denotes the set of requests assigned to vehicle k .

Constraint 2: Total Travel Distance Constraint

The cumulative travel distance under the shared-ride scenario should not exceed that of individual (non-shared) trips:

$$\sum_{k=1}^K \sum_{(i,j) \in E} d_{ij} \cdot x_{ij}^k \leq \sum_{k=1}^K \sum_{(i,j) \in E} d_{ij} \cdot x_{ij}^{k'} \quad (3)$$

where $x_{ij}^k = 1$ if vehicle k traverses edge e_{ij} in the shared ride setting, and $x_{ij}^{k'} = 1$ under the non-shared scenario.

Constraint 3: Pickup Assignment Constraint

Each passenger request must be served exactly once by one vehicle:

$$\sum_{k=1}^K y_r^k = 1, \quad \forall r \in R \quad (4)$$

where $y_r^k = 1$ if vehicle k is assigned to pick up request r , and 0 otherwise.

This formulation integrates both the spatial distribution of demands on edges and the operational constraints of the vehicle fleet. It supports fine-grained route planning in complex urban road networks and serves as a foundation for evolutionary optimization approaches.

4. Improved Genetic Algorithm Design

To enhance the performance and robustness of multi-vehicle route planning, we propose the IGA framework featuring tailored chromosome encoding, a multi-objective fitness function, and a suite of convergence-enhancing strategies.

4.1. The Basic Framework of Genetic Algorithms

GA solve the optimal path allocation problem by simulating the process of natural selection. The algorithm proceeds through several key phases: population initialization, fitness evaluation, selection, crossover, and mutation. By iteratively evolving the population across generations, GA gradually improves the quality of solutions and converges toward a globally optimal multi-vehicle route planning scheme.

GA are adopted in this work due to their natural compatibility with the structure of multi-vehicle path planning problems. The chromosome-based representation allows each road segment, pickup task, or vehicle assignment to be explicitly encoded as a gene, which aligns well with the physical characteristics of urban road networks. This structural expressiveness facilitates the modeling of complex task allocations, capacity constraints, and inter-vehicle coordination in a unified and interpretable form. Compared with other heuristics such as ACO [23] or PSO [24], GA offers greater flexibility in encoding, clearer semantics in crossover operations, and stronger support for preserving feasible path structures—making it particularly suitable for scalable and constraint-aware path planning in intelligent transportation systems. While machine learning-based solvers such as multi-layer perceptrons (MLPs) or reinforcement learning (RL) have shown promise in vehicle routing and dispatching tasks, they typically require extensive training data and computational resources to achieve robust generalization across different network topologies. In contrast, the proposed GA-based framework is a data-efficient, model-free approach that can directly operate on arbitrary road networks without pretraining. This property makes GA especially advantageous for dynamic, large-scale, and constraint-rich urban ride-hailing scenarios where training data may be scarce or rapidly outdated.

In this framework, each chromosome encodes a task allocation scheme for K vehicles, where genes correspond to passenger-carrying road edges rather than nodes. This edge-based representation reflects the reality that passenger pickup occurs along roadways, not merely at intersections. The complete chromosome is represented as a sequence of road segment indices, partitioned into K sub-paths, each corresponding to a vehicle route.

To maintain feasibility, individuals that violate capacity or duplication constraints are either repaired or penalized during fitness evaluation. The fitness function is composed of the total travel distance, the balance of workload among vehicles, and any penalty for capacity overflow. The overall process is summarized in Algorithm 1.

Algorithm 1: Improved Genetic Algorithm with Repair Mechanism

Input: Graph $N = (V, E)$, pickup edges E_d , vehicle count K , vehicle capacity M , population size P , max generations $N_{\max}$

Output: Best chromosome c^* minimizing route cost under constraints

- 1 Initialize population $\mathcal{P}$ with P chromosomes (each encoding a partition of E_d into K feasible sub-paths);
- 2 **for** $g = 1$ to $N_{\max}$ **do**
- 3 **foreach** chromosome $c \in \mathcal{P}$ **do**
- 4 Decode c into vehicle routes $[E_1, \dots, E_K]$;
- 5 // Repair duplicates and ensure feasibility
- 6 Initialize assigned list $\mathcal{A} = \emptyset$;
- 7 **foreach** vehicle route E_k **do**
- 8 **foreach** pickup edge $e \in E_k$ **do**
- 9 **if** $e \in \mathcal{A}$ **then**
- 10 Remove e from E_k ; // Remove duplicate
- 11 **end**
- 12 **else**
- 13 Add e to $\mathcal{A}$;
- 14 **end**
- 15 **end**
- 16 **end**
- 17 Identify unassigned pickup edges $U = E_d - \mathcal{A}$;
- 18 **foreach** $e \in U$ **do**
- 19 Find vehicle v with available capacity and minimal additional route cost;
- 20 Insert e into v 's route at best position;
- 21 **end**
- 22 **foreach** vehicle route E_k **do**
- 23 **while** $\text{Load}(E_k) > M$ **do**
- 24 Select pickup edge e with smallest reassignment cost;
- 25 Move e to another vehicle v with available capacity;
- 26 **end**
- 27 **end**
- 28 Evaluate fitness based on total route cost and residual penalties;
- 29 **end**
- 30 Select parents via Rank with Adjusted Weights;
- 31 Apply boundary-aligned edge-level crossover;
- 32 Apply supervised mutation:
 1. Single-point mutation or random swap of two pickup edges.
 2. After mutation, use a supervised repair process to check for duplicate or missing tasks. Any duplicates are removed, and missing tasks are inserted following the same repair strategy as above.
- Update population using elitism and offspring replacement;
- if** termination criterion met **then**
- break**;
- end**
- 33 **end**
- 34 **return** Best solution c^* with minimal penalized cost

4.2. Design of the Fitness Function

The fitness function guides the evolutionary process by evaluating each chromosome based on routing efficiency and constraint satisfaction. Given that passenger tasks are now modeled on edges, we define the fitness of a solution as the sum of edge traversal costs across all vehicle routes, plus penalties for capacity violations.

In this study, the fitness function is designed with two primary objectives:

- Minimize the total travel distance across all vehicle routes, with a particular emphasis on reducing the length of the longest sub-path to ensure task balance and improve overall system efficiency.
- Enforce strict adherence to the vehicle capacity constraint to avoid overloading during passenger pickup.

Let K denote the number of vehicles, and $E_k = \{e_1, e_2, \dots, e_m\}$ denote the sequence of road edges assigned to vehicle k . Each edge e_{ij} has an associated traversal distance d_{ij} (computed using preprocessed shortest paths on the road network).

The fitness function is formulated as:

$$\text{Fitness} = \sum_{k=1}^K \sum_{e_{ij} \in E_k} d_{ij} + \sum_{k=1}^K \delta_k \quad (5)$$

During the decoding process, the algorithm ensures that each vehicle's task sequence is constructed by connecting consecutive nodes, and the shortest paths between these nodes are computed using Dijkstra's algorithm on the road network graph. This approach captures realistic road connectivity and provides accurate cost estimation.

Where δ_k represents the penalty for vehicle k exceeding its capacity, defined as:

$$\delta_k = \begin{cases} 0, & \text{if } \sum_{e_{ij} \in E_k} m_{ij} \leq M \\ \beta \cdot (\sum_{e_{ij} \in E_k} m_{ij} - M), & \text{otherwise} \end{cases} \quad (6)$$

Here, m_{ij} is the number of passengers waiting on edge e_{ij} , M is the maximum capacity per vehicle (fixed at 4), and β is a penalty coefficient that penalizes overloads. This design encourages feasible, balanced route solutions by simultaneously minimizing travel cost and avoiding capacity violations.

To analyze the impact of the penalty coefficient β on the performance of the proposed algorithm, a sensitivity study was carried out. As summarized in Figure 3, β significantly influences both solution feasibility and overall fitness. When β is set to a relatively low value ($\beta \leq 110$), the algorithm consistently fails to generate feasible solutions, indicating that the penalty is insufficient to effectively mitigate capacity violations. On the other hand, when β becomes excessively large ($\beta \geq 220$), the optimization process suffers from overly aggressive penalization, leading to premature convergence and degraded solution quality. In the intermediate range of $120 \leq \beta \leq 210$, feasible solutions are reliably obtained, with the best fitness value observed at $\beta = 160$. Considering this trade-off between feasibility and convergence, $\beta = 160$ was adopted in the subsequent experiments to ensure both solution validity and optimization effectiveness.

To achieve effective segment connectivity within chromosomes, each gene encodes a pickup edge, and a complete chromosome represents a task allocation plan for all vehicles. When evaluating the fitness of a chromosome, the algorithm first decodes it into K vehicle-specific sub-paths and checks each for validity. If the sub-path violates capacity constraints, a penalty is applied. Then, for each consecutive pair of pickup edges in the sub-path, the shortest route is calculated and accumulated.

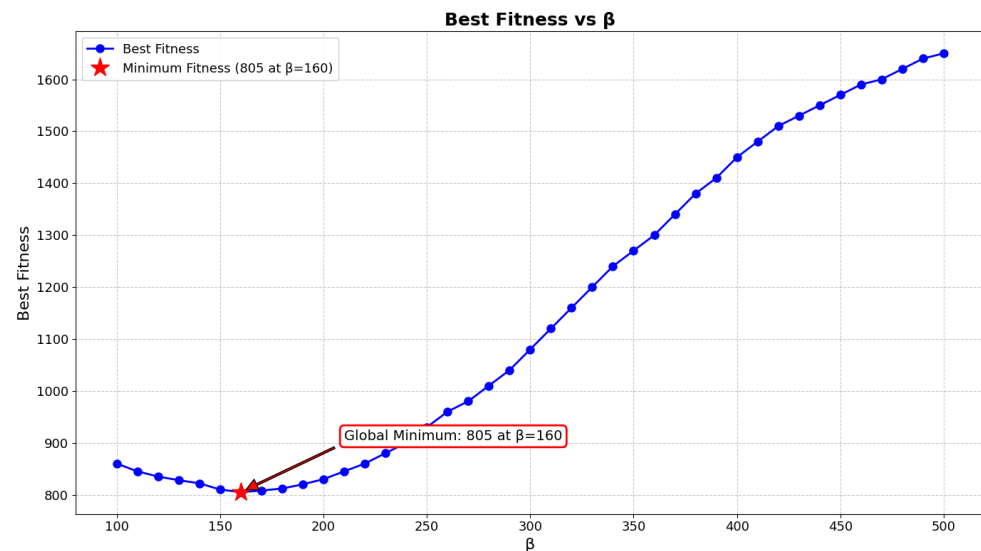


Figure 3. Sensitivity analysis of the penalty coefficient β (with β varying from 100 to 500 in increments of 10). The figure shows the impact of different β values on the convergence behavior and feasibility of solutions. An optimal trade-off is observed at $\beta = 160$, where feasible solutions are consistently obtained with the best fitness performance.

4.3. Chromosome Encoding and Genetic Operations

To effectively represent multi-vehicle pickup tasks on an edge-based road network, each chromosome is designed as a linear sequence of genes, where each gene corresponds to a specific road segment. To enhance clarity, we additionally provide a concrete encoding example in Figure 4, with each sub-sequence encoding the route of a specific vehicle. To enable task segmentation and ensure path planning feasibility, as illustrated in Figure 4, genes are categorized into three distinct types based on their functional roles in the encoding structure [25]:

- **Blue genes :** Represent standard road edges that do not involve passenger pickups. These edges are included in the route to maintain connectivity but are not directly associated with service demands.
- **Yellow genes:** Represent pickup edges, i.e., road edges where passengers are waiting to be picked up. Each pickup edge has an associated demand and must be assigned to exactly one vehicle.
- **Orange genes:** Denote boundary markers that divide the chromosome into sub-sequences, with each sub-sequence encoding the route of a specific vehicle. These boundary genes are critical for partitioning tasks among multiple vehicles.

At the initialization stage, a feasibility-aware sampling strategy is applied to ensure that each pickup edge appears exactly once within a chromosome. Orange boundary genes are then inserted to split the gene sequence into K vehicle-specific sub-paths, forming complete route assignments. This structured representation facilitates subsequent genetic operations, including crossover and mutation, while maintaining task feasibility and supporting convergence toward high-quality solutions.

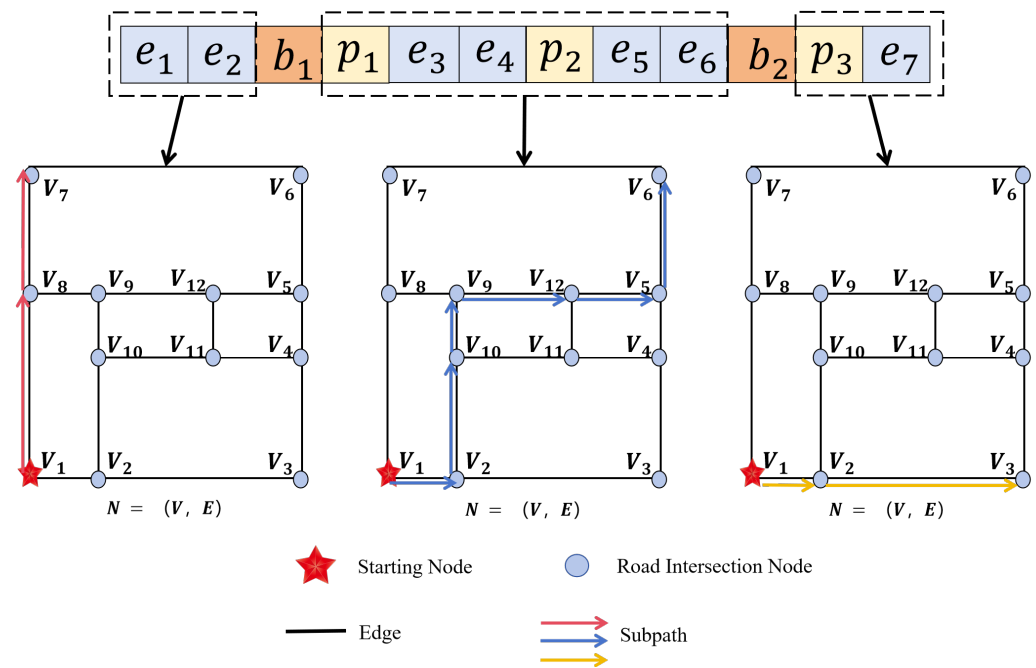


Figure 4. Chromosome encoding for multi-vehicle path planning. Each chromosome consists of three types of genes: blue (standard road edges), yellow (pickup edges), and orange (boundary markers). The illustrated example shows a complete chromosome representation, with each sub-sequence encoding the route of a specific vehicle, separated by orange boundary markers.

A central modeling constraint in this problem is that each pickup edge (yellow gene) must be visited exactly once across the entire fleet. If a chromosome contains duplicate pickup edges—i.e., multiple vehicles are assigned the same task—the individual is considered infeasible. As illustrated in Figure 5, such violations may occur when crossover operations are performed without constraint handling. In this example, the pickup edge p_1 is present in the preserved segment of Parent1 and also inherited from the crossover segment of Parent2, resulting in a duplicated p_1 in Child2.

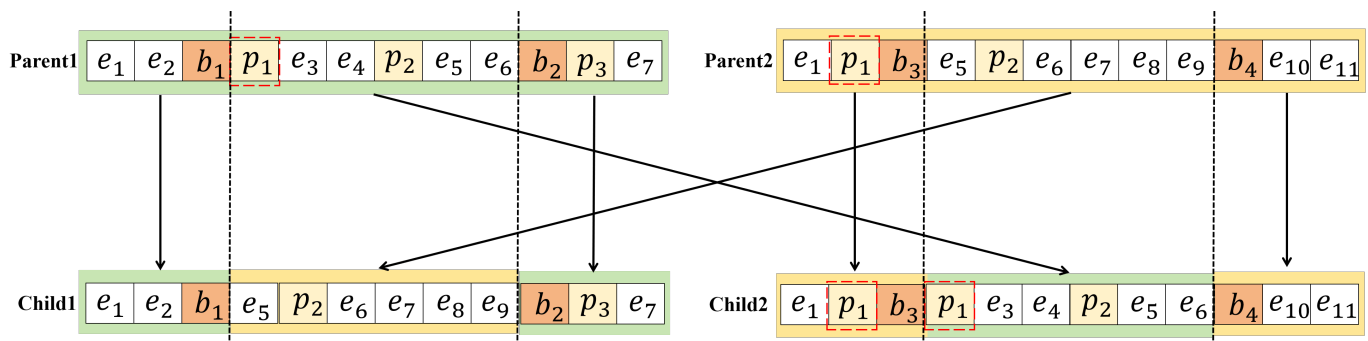


Figure 5. Example of invalid offspring generated by unconstrained crossover. Genes inherited from Parent1 are highlighted in green, while genes inherited from Parent2 are highlighted in yellow. In this example, the pickup edge p_1 appears in both the retained and inherited edges of Child2, violating the uniqueness constraint. The red dashed box indicates the duplicated gene p_1 .

To prevent such conflicts, we employ an edge-level crossover strategy in which crossover points are aligned with predefined boundary genes. This ensures that exchanges occur only between complete vehicle subtours. After crossover, a dedicated repair mechanism validates the offspring by detecting duplicated or missing pickup edges. Duplicates are removed by retaining only the first occurrence, and unassigned tasks are reinserted into feasible positions to restore coverage completeness without exceeding vehicle capacities.

Two mutation operators are applied to maintain diversity and refine route structures. The edge-swap mutation randomly exchanges two pickup edges across subtours, enabling task reassignment between vehicles. The segment inversion mutation reverses a subsequence within a vehicle's route, promoting local structural variation. To ensure feasibility after mutation, a supervised validation process is invoked, checking that all pickup edges are unique and reassigning any lost or duplicated tasks.

During evolution, a multi-objective fitness function guides the search by jointly minimizing the total travel distance and penalizing violations of vehicle capacity constraints. An elitism strategy preserves top-performing individuals across generations, while tournament-based parent selection ensures a balance between selection pressure and exploration. To further enhance selection diversity and control selective pressure, the proposed method incorporates a Rank with Adjusted Weights strategy. Specifically, after ranking individuals by fitness, their selection weights are nonlinearly adjusted using a power-law function:

$$w_i = (\text{rank}_i)^\gamma, \quad (7)$$

where rank_i denotes the ranking position of the i -th individual (higher ranks correspond to better fitness), and γ is an exponent parameter controlling the degree of nonlinearity. When $\gamma > 1$, high-ranking individuals receive disproportionately larger weights, thereby increasing their likelihood of selection. When $\gamma = 1$, the weight grows linearly with rank, while $\gamma < 1$ produces a more uniform distribution that reduces the dominance of top individuals. The power-law function was chosen because it is monotonic, avoids negative weights, requires no complex normalization, and incurs minimal computational overhead, making it suitable for iterative genetic algorithms with large population sizes. In our experiments, we set $\gamma = 1.5$, which provided a favorable trade-off between exploitation of high-quality solutions and maintaining genetic diversity. Mutation rates are also dynamically adjusted—higher in early generations to encourage broad search and gradually reduced in later stages to promote convergence stability.

To conclude the search efficiently, a dual termination criterion is applied: the algorithm halts when either the maximum number of generations is reached or the fitness improvement across successive generations falls below a predefined threshold. This prevents unnecessary prolongation and avoids premature stagnation.

In summary, the proposed genetic algorithm integrates structure-aware encoding, boundary-aligned crossover, supervised mutation, and constraint-driven repair to ensure feasible and efficient evolution. These mechanisms collectively enable the algorithm to generate load-balanced, capacity-compliant, and cost-effective solutions for the multi-vehicle edge-based pickup path planning problem.

5. Experiments and Analysis of Results

To evaluate the performance and generality of the proposed genetic algorithm, we conducted a series of experiments using small-scale road network maps from the Carla simulator. The test environments included the Town02, Town03, Town05 and Town10 maps, as shown in Figure 6.

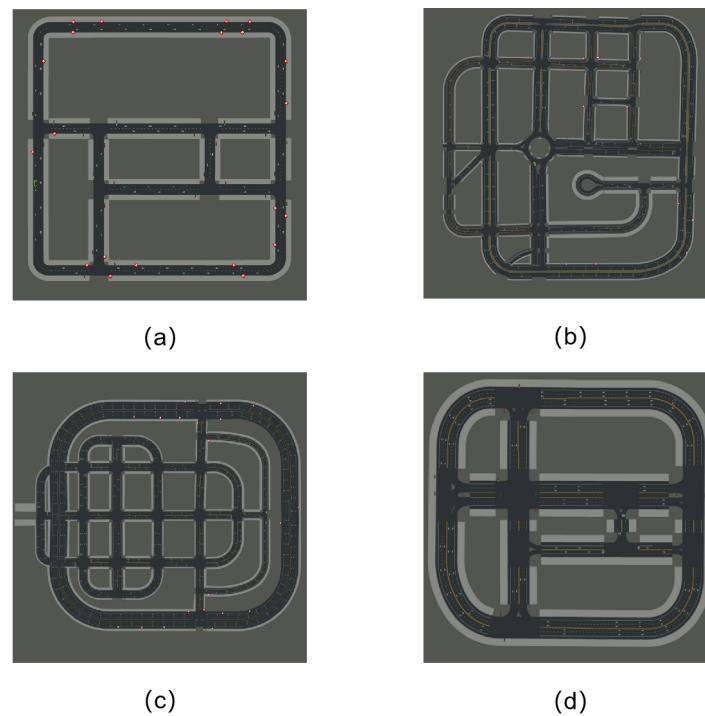


Figure 6. Overview of test maps in CARLA. (a) Town02, (b) Town03, (c) Town05, (d) Town10. The maps represent diverse urban driving environments with variations in road networks, traffic dynamics, and architectural structures, enabling comprehensive testing for autonomous driving systems.

5.1. Experimental Setup

To verify the adaptability of our algorithm to maps of varying sizes and complexities, we selected multiple test environments that differ in structural characteristics such as the number of nodes, edges, and loop closure edges. These parameters are inherently determined by the underlying map topology. Additionally, different levels of passenger demand were configured in each environment to assess the algorithm's performance under diverse task loads. The configuration details are summarized in Table 1.

Table 1. Test Environment Configuration.

Map	Map Size (m)	Number of Nodes	Number of Edges	Loop Closure Edges	Network Length (m)	Number of Passengers
Carla Town02	150 × 150	12	16	3	832	10
Carla Town03	250 × 250	18	24	5	1450	15
Carla Town05	350 × 350	22	32	6	2100	20
Carla Town10	170 × 200	13	18	3	1533	11

Environment configuration details for experimental evaluation. Data are estimated based on CARLA 0.9.14 default maps.

All experiments were conducted under the same control parameters: three vehicles with a maximum passenger capacity of 4 each, a population size of 50, evolution through 1000 generations, a mutation rate of 10%, and 3 elite individuals preserved per generation.

5.2. Experimental Design, Results, and Analysis

5.2.1. Comparison of Selection Strategies in Genetic Algorithm

To assess the influence of different selection strategies on optimization performance, we implemented and compared the following methods:

- **Random Selection:** This method selects individuals completely at random, irrespective of their fitness values. While it does not exploit any information about solution quality, it is occasionally used to maintain genetic diversity or as a baseline for comparative studies [26].
- **Rank-based Selection:** Individuals are ranked according to their fitness, and selection probabilities are assigned based on rank rather than absolute fitness [27]. This approach mitigates the issues associated with highly skewed fitness distributions and helps maintain selection pressure across generations.
- **Rank with Adjusted Weights Selection:** An extension of rank-based selection, this method assigns non-linear or custom-defined weights to different ranks, such as exponential or quadratic scaling. It enables fine-grained control over selection pressure and can be tailored to balance exploration and exploitation.
- **Roulette Wheel Selection:** Also known as fitness-proportional selection, this method allocates selection probability to each individual proportional to its fitness value. It is conceptually simple but can be sensitive to fitness variance, potentially causing premature convergence if a few individuals dominate the population [28].
- **Tournament Selection:** A subset of individuals is randomly sampled from the population to form a "tournament," and the individual with the highest fitness in the group is selected. The size of the tournament determines the selection pressure: larger tournaments favor fitter individuals more strongly.
- **Truncation Selection:** This method involves selecting the top-performing individuals—based on a predefined proportion or fixed number—and discarding the rest. Although it imposes strong selection pressure and accelerates convergence, it may also reduce genetic diversity and increase the risk of premature convergence [29].

All strategies were applied under consistent experimental conditions. The resulting fitness performance across the generations is shown in Figure 7.

From the experimental results, it is clear that the choice of selection strategy plays a critical role in determining both the convergence rate and the quality of the final solution in the genetic algorithm. Table 2 summarizes the best fitness values obtained under each selection strategy.

Table 2. Comparison of Selection Strategies Based on Final Best Fitness.

Selection Strategy	Final Best Fitness
Random Selection	1380
Rank-based Selection	1910
Rank with Adjusted Weights Selection	1170
Roulette Wheel Selection	1840
Tournament Selection	1670
Truncation Selection	2200

Among all tested methods, the Rank with Adjusted Weights strategy consistently outperformed the others, yielding the lowest final fitness value of 1170. This indicates a more effective convergence toward optimal or near-optimal solutions. The improved performance can be attributed to its capacity to strike a better balance between exploration and exploitation, thereby enhancing the algorithm's ability to escape local optima and maintain solution diversity during evolution.

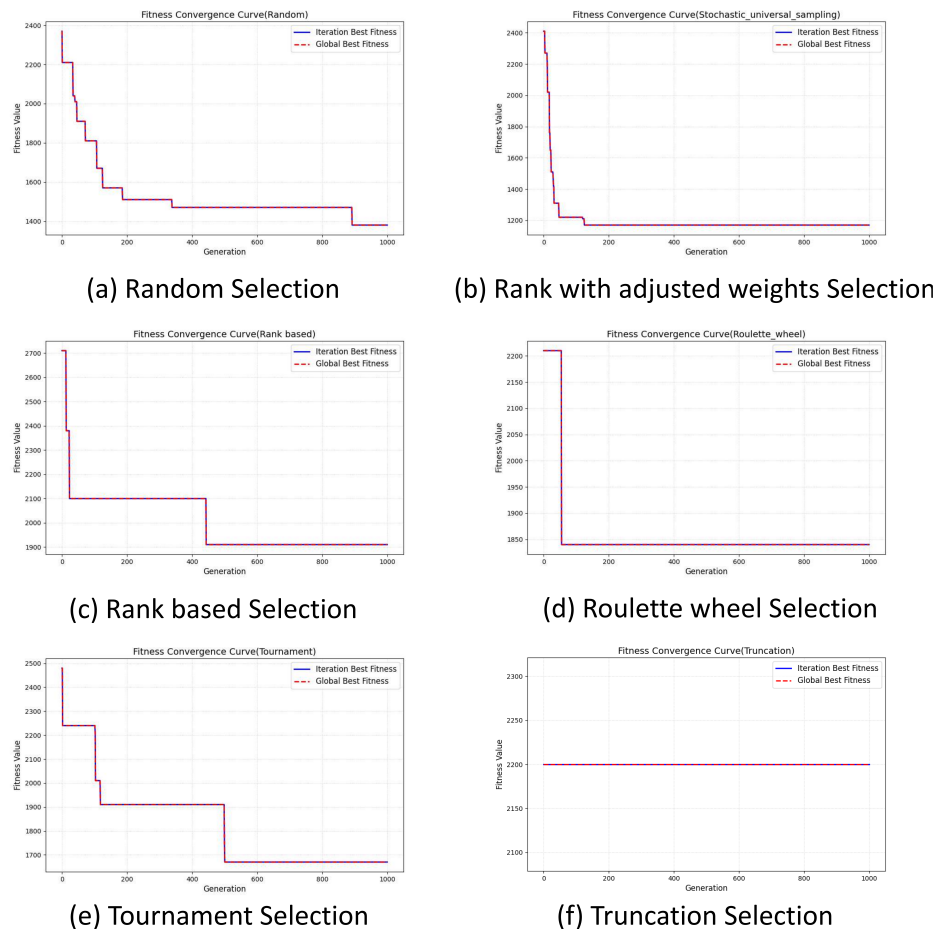


Figure 7. Performance comparison of selection strategies in the genetic algorithm. The Rank with Adjusted Weights strategy (b) achieves the lowest fitness (1170) and fastest convergence, demonstrating superior balance between exploration and exploitation. Rank-based Selection (c) reaches a fitness of 1910, Roulette Wheel Selection (d) achieves 1840, and Tournament Selection (e) obtains 1670. Truncation Selection (f) yields the worst fitness (2200), while Random Selection (a) exhibits slow convergence. Runtime analysis (Table 3) further confirms the efficiency of Rank with Adjusted Weights, with a 20–30% speed advantage over other methods.

Table 3. Algorithm Runtime under Different Selection Strategies.

Selection Strategy	Runtime Mean (s)	Std (s)	95% CI (s)
Random Selection	21.36	0.82	[20.96, 21.75]
Rank with Adjusted Weights Selection	18.77	0.64	[18.45, 19.09]
Rank-based Selection	19.87	0.75	[19.49, 20.25]
Roulette Wheel Selection	22.04	0.89	[21.61, 22.47]
Tournament Selection	22.84	0.91	[22.40, 23.28]
Truncation Selection	22.47	0.88	[22.04, 22.90]

5.2.2. Running Time Analysis

In addition to fitness performance, we also evaluated the algorithm's computational efficiency under different selection schemes. Each runtime value was obtained by averaging 20 independent runs, and we additionally report the standard deviation (Std) to capture variability across runs. For completeness, 95% confidence intervals (CI) were also calculated using Student's *t*-distribution. The results are presented in Table 3.

The Rank with Adjusted Weights Selection strategy not only outperformed in terms of fitness but also recorded the shortest runtime—about 20–30% faster than other strategies. The reported improvements are statistically robust, as indicated by relatively small standard deviations and non-overlapping confidence intervals compared to most other selection

strategies. This efficiency gain highlights its suitability for large-scale or real-time multi-vehicle path planning scenarios. The core idea of this algorithm is to select individuals based on their rankings instead of their original fitness values, and to adjust the selection probability via a power function, which increases the selection pressure for high-ranked individuals and accelerates convergence.

5.2.3. Single-Vehicle vs. Multi-Vehicle Performance Comparison

To further assess the effectiveness of our approach, we conducted experiments comparing the performance of single-vehicle and multi-vehicle path planning using IGA. The tests were carried out on the Carla Town2 and Town10 maps.

For the single-vehicle case, the optimization problem and algorithmic framework remain fully consistent with the multi-vehicle formulation. The change is that the number of vehicles is set to $K = 1$, meaning a single route must cover all required pickup edges. No additional heuristics or solvers were introduced, ensuring a fair and direct comparison with the multi-vehicle setting.

In contrast, the multi-vehicle path planning model based on IGA demonstrated superior global optimization capabilities and better task allocation in complex cooperative search and rescue scenarios. The convergence process of the algorithm for both single- and multi-vehicle settings is illustrated in Figure 8.

As shown in the results, the IGA significantly enhances optimization efficiency in the multi-vehicle case. Specifically, it achieves a 48.3% improvement in average fitness and reduces the number of generations required for convergence by 54.7%, compared to the single-vehicle configuration. This improvement is mainly due to distributing tasks among multiple vehicles, which shortens the maximum and average subpath lengths while maintaining the same edge coverage requirement. This demonstrates that the proposed approach is not only more efficient but also scales better for collaborative path planning problems.

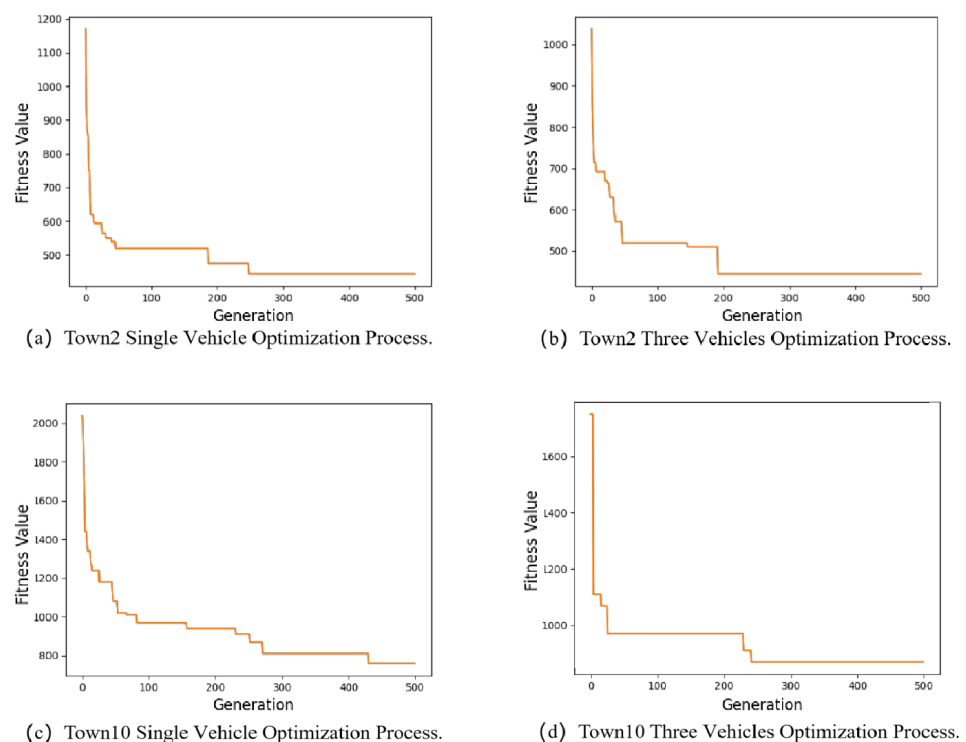


Figure 8. Convergence curves of IGA in single-vehicle ($K = 1$) vs. multi-vehicle scenarios on Town2 and Town10 maps. (a,c) correspond to single-vehicle cases, while (b,d) show multi-vehicle results. The proposed method achieves 48.3% better average fitness and 54.7% faster convergence when multiple vehicles are used, demonstrating improved scalability and efficiency for collaborative path planning.

5.3. Comparison with MSTC*

The Minimum Spanning Tree Control (MSTC*) algorithm represents a classical graph-theoretic method for multi-vehicle path planning. It operates by first generating a Minimum Spanning Tree (MST) to connect all target nodes with minimal cumulative edge cost under static constraints. Subsequently, this tree guides the assignment of traversal paths to vehicles, with the aim of ensuring complete coverage with minimized redundancy. MSTC* is particularly effective in scenarios that demand strong connectivity and global path cohesion.

The MSTC* framework typically comprises two key stages:

- **Minimum Spanning Tree Construction:** An MST is constructed on the weighted graph to establish a globally optimal connectivity structure that minimizes the sum of edge weights.
- **Path Allocation:** Individual subpaths are derived from the MST and distributed among vehicles such that all nodes are visited with minimal overlap and improved efficiency.

To evaluate the effectiveness of IGA, extensive comparative experiments were conducted on four benchmark scenarios: Carla Town02, Town03, Town05, and Town10. Three performance indicators were selected for quantitative analysis: (1) maximum subpath length (Max), (2) average subpath length (Avg), and (3) total runtime (Time). All reported values represent the average performance over 20 independent runs to ensure statistical reliability. In addition to MSTC*, we introduced a standard GA baseline without the proposed edge-based encoding and crossover mechanisms. This classic GA uses a traditional node-based chromosome representation and standard one-point crossover, allowing us to isolate the performance gains specifically attributable to the enhancements introduced in IGA. Table 4 presents the numerical results and percentage improvements of IGA relative to MSTC* and the standard GA.

Table 4. Performance Comparison between IGA, Standard GA, and MSTC* Algorithm. The downward arrow symbol indicates the reduction relative to MSTC*.

Algorithm	Town	Max (m)	Avg (m)	Time (s)	↓ Max (%)	↓ Avg (%)	↓ Time (%)
IGA	Town02	746	740.06	13.09	49.2	43.3	28.5
	Town03	1595	1566.67	42.62	24.0	19.7	12.6
	Town05	2195	2181.67	44.22	23.0	17.7	15.4
	Town10	892	883.06	18.77	48.3	39.3	10.8
GA	Town02	1100	980.50	16.75	25.1	23.6	7.6
	Town03	1830	1750.20	45.20	12.9	9.2	7.3
	Town05	2430	2360.15	48.10	14.0	11.4	8.0
	Town10	1240	1125.30	19.95	28.1	22.6	5.2
MSTC*	Town02	1468	1304.60	18.32	-	-	-
	Town03	2100	1950.25	48.75	-	-	-
	Town05	2850	2650.40	52.30	-	-	-
	Town10	1724.2	1454.30	21.05	-	-	-

5.3.1. Quantitative Analysis and Comparative Results

To validate the effectiveness of the proposed IGA, we conducted a comparative evaluation with the MSTC* method across four representative simulation environments (Town02, Town03, Town05, and Town10). The comparison focused on two core aspects: path optimization performance and computational efficiency.

In terms of path optimization, IGA demonstrated consistently better performance by reducing both the maximum and average subpath lengths assigned to individual vehicles. For example, in the Town02 scenario, the maximum subpath length was reduced from

1468 meters under MSTC* to 746 meters under IGA, representing a 49.2% improvement. Similar reductions were observed in Town10 with a 48.3% decrease, Town03 with 24.0%, and Town05 with 23.0%, further indicating that IGA achieves a more balanced task allocation among vehicles. The average subpath lengths also exhibited marked improvements. In Town02, the average length per vehicle dropped from 1304.60 meters to 740.06 meters, with Town10, Town03, and Town05 showing corresponding reductions of 39.3%, 19.7%, and 17.7%. These results confirm that IGA not only minimizes overall travel cost but also alleviates load imbalance across the vehicle fleet.

From the perspective of computational efficiency, the proposed IGA demonstrated reduced execution times, challenging the common assumption that evolutionary algorithms are inherently time-consuming. In the Town02 environment, runtime decreased from 18.32 s to 13.09 s, reflecting a 28.5% improvement. Similar reductions were observed in Town03, Town05, and Town10, with execution times shortened by 12.6%, 15.4%, and 10.8% respectively. These results highlight the scalability and computational efficiency of the proposed method across varying levels of environmental complexity.

5.3.2. Algorithmic Analysis

The superior performance of IGA can be attributed to a combination of design choices that jointly enhance solution quality and algorithmic efficiency. The use of a multi-objective fitness function that simultaneously considers total route cost and the longest individual subpath allows the algorithm to strike a balance between global optimization and local fairness. Additionally, Dijkstra-based precomputation of pairwise shortest paths enables rapid and accurate evaluation of solution quality, which contributes to faster convergence. The power-adjusted selection mechanism further enhances search pressure while preserving population diversity, and the edge-level crossover strategy ensures that offspring remain structurally valid and feasible, mitigating the risk of premature convergence.

From a practical perspective, these improvements translate into shorter overall routing distances, faster computation times, and more balanced task allocation across vehicles. By effectively minimizing the longest individual subpath, IGA prevents excessive workload concentration on specific vehicles, improving operational fairness and service efficiency. The algorithm's ability to rapidly converge to high-quality, feasible solutions also makes it more suitable for time-sensitive applications, such as on-demand ride-hailing, logistics distribution, and coordinated rescue operations in complex urban traffic networks.

Together, these components enable IGA to effectively address the dual challenges of task allocation balance and computational tractability, making it well-suited for complex multi-vehicle route planning tasks in urban traffic networks.

6. Conclusions and Future Work

This paper presents a novel multi-vehicle collaborative path planning framework based on the IGA, targeting the core challenges of rescue search tasks, including path conflict mitigation, task allocation balance, and adaptability to dynamic obstacles. The proposed method integrates a multi-objective fitness function, Dijkstra-based path estimation, and an edge-level crossover mechanism, which collectively improve convergence speed, solution quality, and environmental adaptability within complex urban settings.

Extensive experiments conducted on benchmark scenarios (Carla Town02, Town03, Town05 and Town10) demonstrate the superior performance of the proposed IGA method compared to the classical MSTC* algorithm. Specifically, the IGA achieves substantial improvements, with reductions of up to 49.2% in maximum subpath length, 43.3% in average subpath length, and 28.6% in runtime. These results validate the feasibility, efficiency, and scalability of the proposed approach for real-world multi-agent planning applications.

Beyond ride-hailing scenarios, the proposed IGA-based framework can also be applied to other domains that require coordinated multi-vehicle routing under constraints, such as urban logistics distribution, last-mile delivery, and collaborative search-and-rescue operations. Recent advances in machine learning, reinforcement learning (RL) [30], and graph neural networks (GNNs) have also demonstrated promising results in solving complex routing problems. Integrating these learning-based approaches into our IGA framework offers an exciting opportunity to further improve decision-making, exploit richer spatial-temporal features of traffic networks, and enhance adaptability to dynamic and uncertain environments.

Future work will extend the current research in several directions to further enhance its applicability within intelligent transportation systems. A primary focus will be on improving real-time performance, enabling the proposed IGA framework to make rapid and reliable path planning decisions in dynamic, time-sensitive urban environments. This includes optimizing algorithmic efficiency and integrating real-time traffic data and sensor feedback to support adaptive decision-making. Additionally, incorporating learning-based components such as reinforcement learning and graph neural networks will be explored to improve the algorithm's capacity to generalize across complex and previously unseen road network topologies. Bridging the gap between simulation and real-world deployment is also a key direction, involving validation of the method on physical platforms or intelligent fleet management systems to assess robustness and scalability under practical conditions.

Moreover, the proposed model can be naturally extended to handle heterogeneous pickup conditions, such as time-window-constrained requests, dynamically entering or exiting vehicles, and partial shared-ride acceptance, thereby enhancing its adaptability to broader on-demand mobility scenarios. Future research will also explore the integration of Dynamic Movement Primitives (DMPs) and adaptive re-planning mechanisms, enabling vehicles to adjust routes online in response to changing passenger demands and environmental uncertainties. In addition, collision avoidance mechanisms will be incorporated to ensure safe multi-vehicle coordination during execution, and simulated force sensing will be introduced to model passenger interactions and vehicle-environment contacts. These extensions aim to enhance both the safety and realism of the proposed framework when deployed in complex, dynamic urban settings.

These efforts aim to transform the IGA framework into a viable solution for large-scale, coordinated vehicle task allocation and path planning within future intelligent transportation infrastructures.

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